Over 30 million people in India alone are affected by **locomotor disability** (ICRC study, 2015), 1% of a population need a wheelchair (WHO)



Multiple sclerosis,
Amputations, Aging



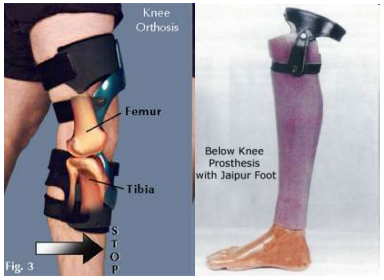
Muscular dystrophy,
Cerebral palsy

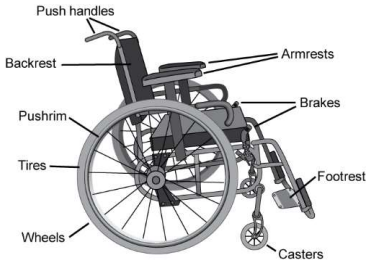


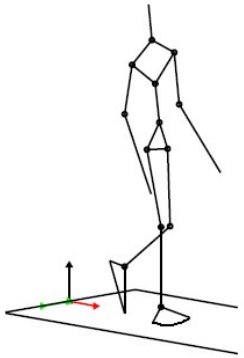
Spina bifida,
Post-polio syndrome



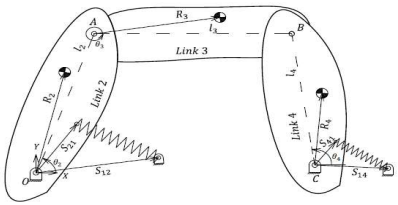
Traumatic
brain / spinal chord injury,
Paraplegia, Quadriplegia



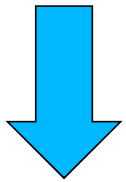


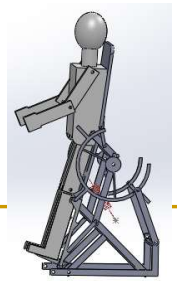


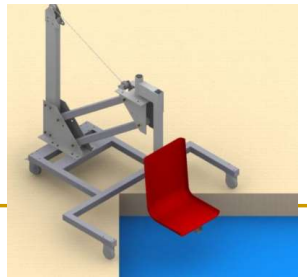
Biomechanics



Mechanisms









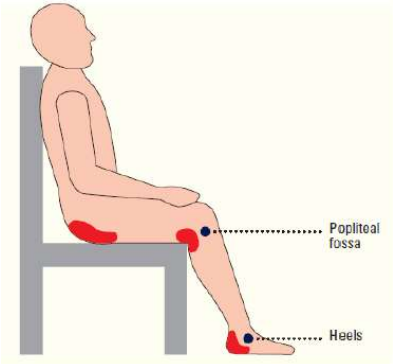
TTK Center for Rehabilitation Research & Device Development (R2D2)
Department of Mechanical Engineering
https://home.iitm.ac.in/r2d2

Need for Standing



Reasons for wheelchair use

Spinal Cord Injury, Brain Injury, Multiple Sclerosis,
Polio, Amputation, Aging, Muscular Dystrophy, Cerebral Palsy



Problems of Prolonged Sitting

Pressure Sores, Contractures, Swelling of legs & feet,
Risk of fractures

Benefits of Standing

Functionality



Self Esteem



Health Benefits

↑
- Circulation
- Bone Health
- Range of Motion
- Vital Organ Functioning

↓
- Pressure Sores
- UTIs
- Spasticity
- Contractures

Current Methods



Orthosis & Crutches

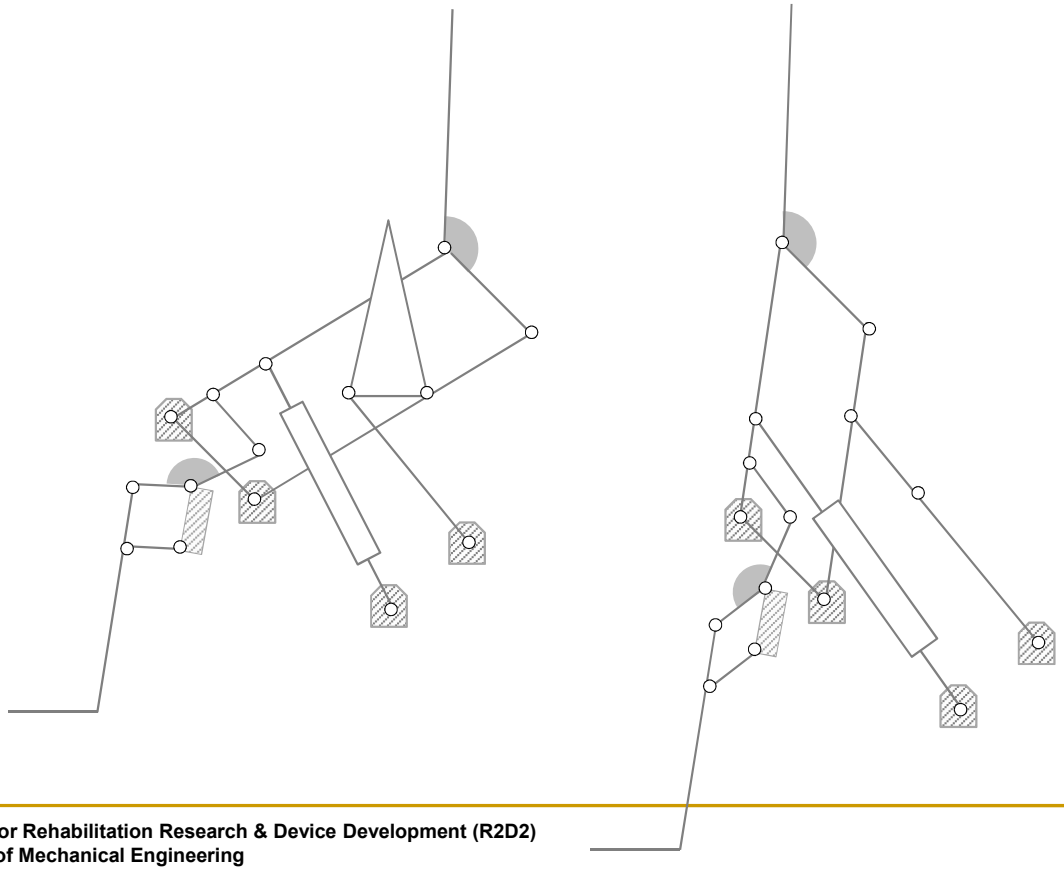
Cumbersome and unsafe process



Standing Frame

Needs assistance from 2-3 people

Standing wheelchair



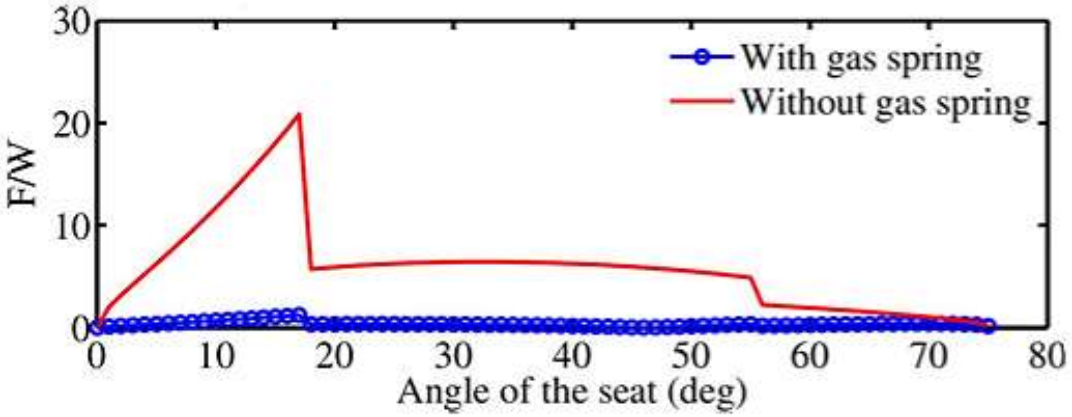


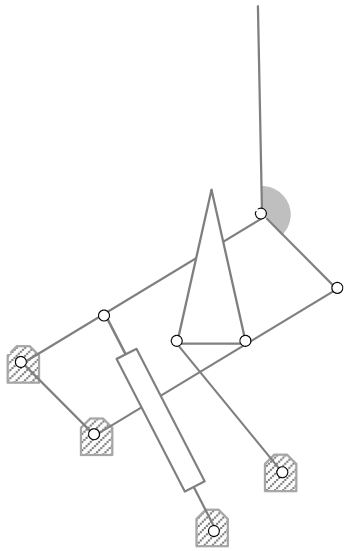
Standing wheelchair

- Hand powered
- Linkage based mechanism
- Handle provided for controlled actuation
- Optimized spring setting reduces effort

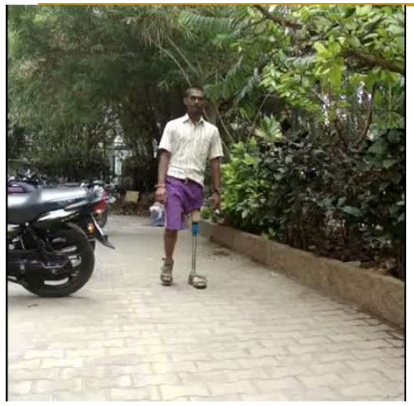


Standing wheelchair





R2D2 Overview





IITM Knee

NeoRider



Pool Lift



Body Movement Wheelchair



NeoBolt



Saathi Walker

